

Subgroup

Let $(G, *)$ be a group and $H \subseteq G$ be a non-empty subset that satisfies the following conditions

- 1) For $a, b \in H$, $a * b \in H$.
- 2) $e \in H$, where e is the identity of $(G, *)$.
- 3) For any $a \in H$, $a^{-1} \in H$ then $(H, *)$ is called a subgroup of $(G, *)$.

Note:

1. $[H, *)$ is itself a group with the same identity as that of $(G, *)$ and with the same binary operation $*$ defined on G .

2. Obviously $(e, *)$ and $(G, *)$ are trivial subgroups of $(G, *)$.

All other subgroups are called proper subgroups.

Example:

1. the additive group of even integers is a subgroup of the additive group of all integers.
2. the multiplicative group $(1, -1)$ is a subgroup of the multiplicative group $\{1, -1, i, -i\}$.

Sub Semigroup

If $(S, *)$ is a semigroup and $T \subseteq S$ is closed under the operation $*$, then $(T, *)$ is called a subsemigroup of $(S, *)$.

Ex:

if the set E of all even non negative integers, then $(E, +)$ is a subsemigroup of the semigroup $(N, +)$, where N is the set of natural numbers.

Let $(N, +)$ be the semigroup & $2N \subseteq N$

$\therefore (2N, +)$ is closed under $+$

$\therefore (2N, +)$ is a subsemigroup.

Let $(N, +)$ be the semigroup

$$\& 2N \subseteq N$$

$\therefore (2N, +)$ is closed under '+'

$\therefore (2N, +)$ is a subsemigroup.

Submonoid:

If $(M, *, e)$ is a monoid,

$T \subseteq M$ is closed under the operation $*$

and $e \in T$, then $(T, *, e)$ is called

a submonoid of $(M, *, e)$.

Ex:

If E is the set of all non-negative even integers, then $(E, +, 0)$ is a submonoid of $(\mathbb{N}, +, 0)$, where $\mathbb{N}$ is the set of natural numbers.

Let $(\mathbb{N}, \cdot)$ be the monoid & $1 \in \mathbb{N}$
odd numbers of $\mathbb{N} \subseteq \mathbb{N}$

$\therefore$ Odd numbers of $\mathbb{N}$ is closed under " $\cdot$ ".

& $1 \in$ odd numbers of $\mathbb{N}$.

$\therefore$ (Odd numbers of $\mathbb{N}, \cdot$) is a Submonoid.

i.e. $(2\mathbb{N}-1, \cdot)$ is a submonoid.

Homomorphism:

Let $(X, \circ)$ and $(Y, *)$ be two algebraic systems where $\circ$ and $*$ are binary (n-ary) operations, then a mapping $g: X \rightarrow Y$ is called a homomorphism or simply morphism from $(X, \circ)$ to $(Y, *)$, if for any $x_1, x_2 \in X$, $g(x_1 \circ x_2) = g(x_1) * g(x_2)$

Group Homomorphism:

Let $(G, *)$ and (H, Δ) be the two groups. A mapping $g: G \rightarrow H$ such that

$$(i) \quad g(a * b) = g(a) \Delta g(b) \quad \text{for any } a, b \in G$$

$$(ii) \quad g(e_G) = e_H \quad \text{for } e_G \in G \text{ and } e_H \in H$$

$$(iii) \quad g(a^{-1}) = (g(a))^{-1} \quad \text{for } a^{-1} \in G \text{ and } (g(a))^{-1} \in H.$$

Then g is said to be a group homomorphism.

Semigroup homomorphism:

Let $(S, *)$ and (T, Δ) be any two semigroups, then a mapping $g: S \rightarrow T$ such that, for any two elements $a, b \in S$,
$$g(a * b) = g(a) \Delta g(b)$$
is called a semigroup homomorphism.

Monoid homomorphism:

If $(M, *, e_M)$ and (T, Δ, e_T) are any two monoids, where e_M and e_T are identity elements

of M and T with respect to the corresponding binary operations $*$ and Δ respectively, then a mapping $g: M \rightarrow T$ such that, for any two elements $a, b \in M$,

$$g(a * b) = g(a) \Delta g(b)$$

$$\text{and } g(e_M) = e_T$$

is called a monoid homomorphism.

Kernel of a homomorphism:

Let $(G, *)$ and (H, Δ) be the two groups and let $g: G \rightarrow H$ such that for any

$$a \in G,$$

$$(i) \quad g(a) = e_H$$

$$(ii) \quad g(e_G) = e_H \quad \text{where } e_G \in G$$
$$e_H \in H$$

Then g is said to be kernel of the homomorphism and it is denoted by $\ker g$.

9) Theorem:

If $f: G \rightarrow G'$ is a group homomorphism from $\{G, *\}$ to $\{G', \Delta\}$, then

(i) $f(e) = e'$, where e and e' are the identity elements of G and G' respectively,

(ii) for any $a \in G$, $f(a^{-1}) = [f(a)]^{-1}$

(iii) if H is a subgroup of G , then

$f(H) = \{f(h) \mid h \in H\}$ is a group of G' .

Proof:

(i) $f(e * e) = f(e) \Delta f(e)$, by definition of homomorphism

$$\text{i.e. } f(e) = f(e) \Delta f(e)$$

i.e. $f(e)$ is an idempotent element of $\{G', \Delta\}$

But the only idempotent element of a group is its identity.

$$\therefore f(e) = e'$$

(ii) For any $a \in G$, $a^{-1} \in G$

$$\therefore f(a * a^{-1}) = f(a) \Delta f(a^{-1})$$

$$\text{i.e. } f(e) = f(a) \Delta f(a^{-1})$$

$$\text{i.e. } e' = f(a) \Delta f(a^{-1})$$

Similarly,

$$f(a^{-1} * a) = f(a^{-1}) \Delta f(a)$$

$$\text{i.e. } e' = f(a^{-1}) \Delta f(a)$$

From (1) & (2), we see that $\rightarrow \textcircled{2}$

$f(a^{-1})$ is the inverse of $f(a)$.

$$\text{i.e. } f(a^{-1}) = (f(a))^{-1}$$

(iii) Let $h_1, h_2 \in H$

Then $h_1' = f(h_1)$ and $h_2' = f(h_2) \in f(H)$

$$\begin{aligned}\text{Now } h_1' \Delta (h_2')^{-1} &= f(h_1) \Delta [f(h_2)]^{-1} \\ &= f(h_1) \Delta f(h_2^{-1}) \quad (\text{by (ii)}) \\ &= f(h_1 * h_2^{-1}) \quad \text{by homomorphism} \\ &= f(h_3) \quad \text{where } h_3 = h_1 * h_2^{-1} \in H,\end{aligned}$$

$$\therefore h_1' \Delta (h_2')^{-1} \in f(H)$$

Thus $h_1', h_2' \in f(H) \Rightarrow h_1' \Delta (h_2')^{-1} \in f(H)$

$\therefore f(H)$ is a subgroup of G' .

① P.T. A non empty subset H of a group G is a subgroup of G iff $a, b \in H \Rightarrow a * b^{-1} \in H$.

(or) The necessary and sufficient condition for a non empty subset H of a group $\{G, *\}$ to be a subgroup is $a, b \in H \Rightarrow a * b^{-1} \in H$.

proof:

Let H be a subgroup of G .

then for $a, b \in H$, $b^{-1} \in H$ such that

$$a * b^{-1} \in H. \quad (\text{by closure property})$$

Thus, the condition is necessary.

Conversely,

Let $a * b^{-1} \in H$ where $a, b \in H$ and

H is a nonempty subset of G .

If $b=a$, the given condition gives

$$a * a^{-1} \in H$$

$$\text{i.e. } e \in H \longrightarrow \textcircled{1}$$

Using the given condition for the pair $e, a \in H$, we have $e * a^{-1} \in H$

$$\text{i.e. } a^{-1} \in H \longrightarrow \textcircled{2}$$

Similarly, $b^{-1} \in H$

Using the given condition for the pair a and $b^{-1} \in H$, we have

$$a * (b^{-1})^{-1} \in H$$

$$\text{i.e. } a * b \in H \longrightarrow \textcircled{3}$$

From (1), (2) & (3), H is a subgroup.

Thus the condition is sufficient.

② P.T. Intersection of two subgroups of G is a subgroup of G but the union of two subgroups of a group need not be a subgroup of G .

Proof:

Let H and K are two subgroups of G .

$\therefore$ For $a, b \in H \& K$

$$\Rightarrow ab^{-1} \in H \& ab^{-1} \in K$$

$$\therefore ab^{-1} \in H \cap K$$

$\therefore H \cap K$ is a subgroup.

Also,

Let $(\mathbb{Z}, +)$ be a group

and $(2\mathbb{Z}, +)$ & $(3\mathbb{Z}, +)$ be the two subgroups of $(\mathbb{Z}, +)$.

$$\text{where } 2\mathbb{Z} = \{0, \pm 2, \pm 4, \dots\}$$

$$3\mathbb{Z} = \{0, \pm 3, \pm 6, \dots\}$$

$$2\mathbb{Z} \cup 3\mathbb{Z} = \{0, \pm 2, \pm 3, \pm 4, \pm 6, \dots\}$$

$$\text{Here } 3+4 = 7 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$$

$\therefore$ Closure fails

$\therefore$ Union of two subgroups need not be a subgroup of a group G .

③ Let $(S, *)$, (T, Δ) & $(V, \oplus)$ be semigroups and $g: S \rightarrow T$ & $h: T \rightarrow V$ be semigroup homomorphisms then $(h \circ g): S \rightarrow V$ is a semigroup homomorphism from $(S, *)$ to $(V, \oplus)$.

proof:

Let $a, b \in S$ then

$$(h \circ g)(a * b) = h(g(a * b))$$

$$= h(g(a) \Delta g(b)) \quad (\because g \text{ is homomorphism})$$

$$= h(g(a) \oplus g(b))$$

$$(h \circ g)(a * b) = (h \circ g)(a) \oplus (h \circ g)(b)$$

$\therefore (h \circ g): S \rightarrow V$ is a semigroup homomorphism.

④ For any Commutative monoid $(M, *)$ the set of idempotent elements of M forms a submonoid.

Proof:

Let S be the set of idempotent elements of M , where $S \subseteq M$.

Since $e * e = e$, e is an idempotent element of M and hence $e \in S$.

Let $a, b \in S$.

Then $a * a = a$ and $b * b = b$.

$$\begin{aligned} \text{Now } (a * b) * (a * b) &= a * (b * a) * b \\ &= a * (a * b) * b \quad (\text{since } M \text{ is commutative}) \\ &= (a * a) * (b * b) \\ &= a * b \end{aligned}$$

Hence $a * b$ is also an idempotent element.

$\therefore a * b \in S$ and $(S, *)$ is a submonoid.

⑤ Let $(S, *)$ be a given semigroup.

$\exists$ a homomorphism $g: S \rightarrow S^S$

where $(S^S, \circ)$ is a semigroup
of function from S to S under
the operation of (left) composition.

proof:

For any element $a \in S$, let $g(a) = f_a$,
where $f_a \in S^S$ is defined by

$$f_a(b) = a * b, \text{ for any } b \in S$$

$\rightarrow \textcircled{6}$

Now $g(a * b) = f_{a * b} \rightarrow (2)$

where $f_{a * b}(c) = (a * b) * c$

$= a * (b * c)$ (by the associativity of the semigroup $(S, *)$)

$= f_a(f_b(c))$ (by (1))

$= (f_a \circ f_b)(c)$

$= (g(a) \cdot g(b))(c)$

$\therefore f_{a * b} = g(a) \cdot g(b) \rightarrow (3)$

From (2) & (3), we get

$g(a * b) = g(a) \cdot g(b)$

$\therefore g: S \rightarrow S^S$ is a homomorphism.

⑥ Let $(S, *)$ & (T, Δ) be two semigroups & g be a semigroup homomorphism from $(S, *)$ to (T, Δ) corresponds to g $\exists$ a Congruence relation R on $(S, *)$ defined by $x R y \iff g(x) = g(y)$ for $x, y \in S$.

Proof:

Let $x_1, x_2, x_1', x_2' \in S$

such that $x_1 R x_1'$ & $x_2 R x_2'$

$$\begin{aligned} \text{from } g(x_1 * x_2) &= g(x_1) \Delta g(x_2) \\ &= g(x_1') \Delta g(x_2') \\ &= g(x_1' * x_2') \end{aligned}$$

$\therefore R$ is a Congruence relation on $(S, *)$.

⑦ P.T. Kernel of a homomorphism g from a group $(G, *)$ to (H, Δ) is a subgroup.

Proof:

We know that

$$\text{Ker}(g) = \{ a \mid a \in G \text{ \& } g(a) = e_H \}$$

where $e_H \in H$
& $a \in G$

Let $g: G \rightarrow H$ be the kernel homomorphism

i) for $a, b \in \text{Ker } g$

To prove: $a * b \in \text{Ker } g$

Since $a, b \in \ker g \Leftrightarrow g(a) = g(b) = e_H$

$$\therefore g(a * b) = g(a) \Delta g(b)$$

$$= e_H \Delta e_H$$

$$= e_H$$

$\therefore a * b \in \ker g$

ii) for $e_G \in G$ then $\exists e_H \in H$

such that $g(e_G) = e_H$

$\therefore e_G \in \ker g$

iii) for $a \in \ker g$

$$g(a^{-1}) = (g(a))^{-1} = e_H^{-1} = e_H$$

$\therefore a^{-1} \in \ker g$

$\therefore \text{Ker } g$ is a subgroup of $(G, *)$
to (H, Δ) .

P.T. $\text{Ker } g$ is a normal subgroup.

for any $a \in G$ & $k \in \text{Ker } g$

$$\begin{aligned}\therefore g(a^{-1} * k * a) &= g(a^{-1}) \Delta g(k) \Delta g(a) \\ &= e_H \Delta e_H \Delta e_H \\ &= e_H\end{aligned}$$

$$\therefore a^{-1} * k * a \in \text{Ker } g$$

$\therefore \text{Ker } g$ is a normal subgroup.